Layer 5: Reference and Counter Electrodes

On-Chip Ag/AgCl Quasi-Reference Stability, Counter Electrode Sizing, and Solution Resistance

Eton iGEM 2026, Dry Lab | Layer 5 of 17-Layer Biosensor Pipeline | `reference_electrode.py` · `counter_electrode.py`

Layer 5 characterises the two supporting electrodes of the three-electrode cell – the on-chip Ag/AgCl quasi-reference (RE) and the counter electrode (CE) – neither of which carries the biochemical signal itself, but both of which set the accuracy and stability of the potential axis on which every downstream SWV measurement depends. RE drift shifts the SWV potential axis and, if severe enough, clips the peak-integration window (Layer 12); an undersized CE limits the current the cell can pass and distorts the SWV waveform. This report follows the Layer 5 content specification (Sec. 5.1–5.5) for the RE Nernst model, drift budget, CE current budget, CE sizing/gas-evolution check, and solution-resistance/iR analysis, and cross-checks each numeric assumption against the confirmed values from the layers that have since been completed downstream (Layers 7, 9, 10, 13).

Architecture note. The content spec (Sec. 5.3) was written against an early planning picture: an 8-working-electrode (WE) cartridge running a dual MMP-9/HNE channel with a wide −0.65 to +0.65 V scan window (README Layer 11 draft). The confirmed downstream design (Layer 13) is a single MMP-9 channel with active/D-enantiomer-control electrode pairs (Layer 13, Sec. III), each carrying a co-deposited ferrocene (C6FC) internal-area reference alongside the MB probe (Layer 7), and a confirmed SWV scan window of −0.65 to +0.10 V vs Ag/AgCl (Layer 7 Eq. III.7; Layer 10 Sec. III) – narrower than originally planned, because only the MB reduction peak plus a nearby Fc oxidation peak need to be captured, not a symmetric ±0.65 V window built around two well-separated channels. Sections III–IV below carry both the original-planning and confirmed numbers side by side, since both remain useful: the planning numbers bound the worst case if the electrode count is later scaled up, and the confirmed numbers describe the as-designed device.

I. On-Chip Reference Electrode Potential

Two Ag/AgCl conventions, and why they must not be conflated

The Nernst equation for the Ag/AgCl couple is

$$E_{\text{ref}} = E^{\circ}_{\text{AgCl/Ag}} - \frac{RT}{F}\ln a_{\text{Cl}^-}, \qquad E^{\circ}_{\text{AgCl/Ag}} = +0.2224 \text{ V vs NHE at } 25^{\circ}\text{C} \tag{I.1}$$

(Bard & Faulkner, *Electrochemical Methods*, 2nd ed., Table A.1 – confirmed standard value). This is *not* the same number as the potential of a conventional saturated-KCl Ag/AgCl reference electrode, $E_{\text{AgCl(sat.)}} = +0.197$ V vs NHE, which is the reference convention used for the literature formal potentials quoted elsewhere in this pipeline (e.g. Layer 10's $E^{\circ\prime}_{\text{MB}} = -290$ mV vs Ag/AgCl, Pheeney & Barton 2012). The two differ because the saturated-KCl electrode operates at $a_{\text{Cl}^-} \approx 1$ inside its own internal filling solution, not at the sample's chloride activity. The on-chip quasi-reference in this device has no internal filling solution – it equilibrates directly with the sample via Eq. (I.1) – so its potential depends on the *sample's* chloride activity and will differ from both of these conventions.

On-chip potential in wound fluid

Wound fluid chloride is 110–120 mM (content spec Sec. 5.1). At physiological ionic strength, the mean activity coefficient for a 1:1 electrolyte is $\gamma \approx 0.75$ **[FLAG]** (Debye-Hückel extended-law estimate; not measured for wound fluid specifically), giving $a_{\mathrm{Cl^-}} \approx 0.083$–$0.090$ and

$$E_{\mathrm{ref}} \approx +0.284 \text{ to } +0.287 \text{ V vs NHE} \qquad (\text{mid } +0.285 \text{ V}) \tag{I.2}$$

(`reference_electrode.py`, Fig. 1A). This sits +87 to +90 mV *positive* of the saturated-KCl reference (+0.197 V vs NHE), i.e. an offset $E_{\mathrm{offset}} \approx +88$ mV (midpoint), consistent with the content spec's own worked estimate of +0.085 V.

Correcting literature peak positions

Because the on-chip electrode sits E_{offset} positive of the saturated reference on the absolute (NHE) scale, the *same* redox couple reads E_{offset} more negative when expressed vs the on-chip quasi-reference than vs the saturated convention:

$$E_{\mathrm{applied,\ on\text{-}chip}} = E_{\mathrm{lit,\ sat.}} - E_{\mathrm{offset}} \tag{I.3}$$

Applying this to the two formal potentials confirmed elsewhere in the pipeline:

Couple	vs sat. Ag/AgCl (lit.)	Correction	vs on-chip quasi-ref
$E^{\circ\prime}_{\mathrm{MB}}$ (Pheeney 2012, Layer 10)	−290 mV	−88 mV	−378 mV
E°_{Fc} range (Layer 10, Sec. VII)	+400 to +470 mV	−88 mV	+312 to +382 mV

Key finding. Layers 7 and 10 state their scan window and formal potentials "vs Ag/AgCl" without specifying whether this is the saturated-reference convention (the source-paper convention) or the on-chip quasi-reference derived here. The two conventions differ by ~ 90 mV – small next to the confirmed scan window (−0.65 to +0.10 V) and MB peak FWHM (65 mV, Layer 11), so neither peak is at risk of falling outside the window under either convention, but the ambiguity should be resolved explicitly in Layer 10/11 before the two reports' numbers are combined quantitatively.

II. Reference Electrode Drift

Three drift mechanisms are budgeted (content spec Sec. 5.2), each evaluated in `reference_electrode.py`:

(a) $\mathrm{Cl^-}$ depletion/accumulation at the RE surface. As the assay proceeds, $\mathrm{Cl^-}$ diffuses between the RE surface and bulk. To first order,

$$\Delta E_{\mathrm{ref}} = \frac{RT}{F}\,\Delta \ln a_{\mathrm{Cl^-}} \approx \frac{RT}{F}\cdot\frac{\Delta[\mathrm{Cl^-}]}{[\mathrm{Cl^-}]} \tag{II.1}$$

For a 5% surface concentration change: $\Delta E_{\mathrm{ref}} \approx 1.3$ mV – tolerable.

(b) AgCl dissolution. $K_{sp} = 1.8 \times 10^{-10}$ (confirmed, standard value). At $[\mathrm{Cl^-}] = 115$ mM: $[\mathrm{Ag^+}] = K_{sp}/[\mathrm{Cl^-}] \approx 1.6 \times 10^{-9}$ M – the equilibrium free-silver concentration is negligible, so dissolution does not meaningfully deplete a compact AgCl reservoir over a 30-min assay.

(c) Photosensitivity. Ag/AgCl decomposes under UV/visible light; an illuminated quasi-reference can drift 1–10 mV. This is a design requirement (opaque/dark cartridge), not a physical constant, and is eliminated by construction rather than budgeted as residual error.

Combined budget. Modelling $E_{\text{ref}}(t)$ as a random walk with variance $\propto t$ (Fig. 1B) and calibrating its step size so that the 30-min RMS drift matches the content spec's target (< 2 mV for a sound RE design) gives comfortable margin against the two thresholds that matter: the < 5 mV acceptable total drift (below which the ratiometric peak-*area* calculation is essentially unaffected, since areas are far less sensitive to small position shifts than peak heights), and the 25 mV peak-clipping-risk threshold (one MB peak HWHM, from the 65 mV FWHM confirmed in Layer 11) beyond which the Layer 12 Gaussian-fit integration window could clip the peak and bias the area estimate.

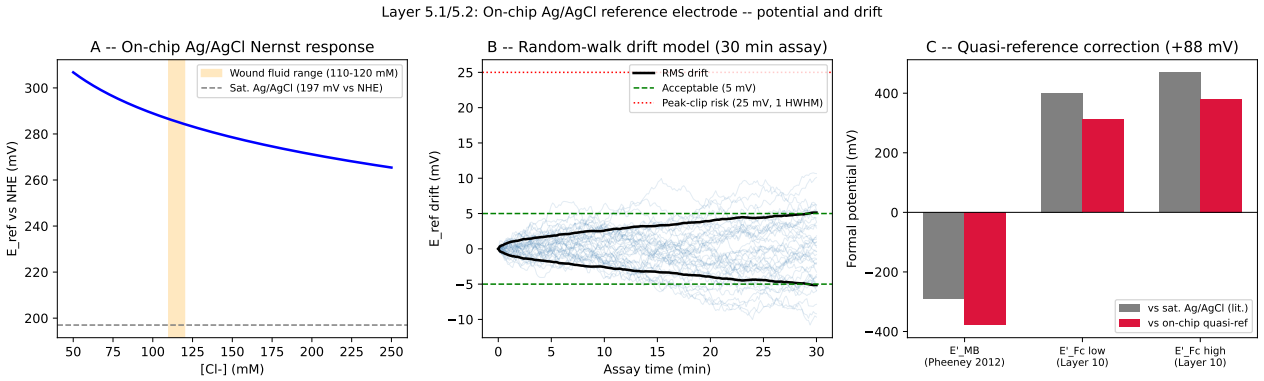


Figure 1: **A**: On-chip Ag/AgCl Nernst response vs [Cl^-], with the wound-fluid range and saturated-reference convention marked. **B**: Random-walk drift model over a 30-min assay, with the acceptable-drift and peak-clipping-risk thresholds overlaid. **C**: Literature formal potentials (vs saturated Ag/AgCl) corrected to the on-chip quasi-reference scale (Eq. I.3).

III. Counter Electrode Current Budget and Sizing

Per-WE current

$$I_{\text{cap}} = 2\, C_{dl}\, A_{\text{WE}}\, \Delta E_{SW}\, f_{SWV} \tag{III.1}$$

Using the Layer 6 mid-range value $C_{dl} = 2\ \mu\text{F/cm}^2$, $A_{\text{WE}} = 1\ \text{mm}^2$, $\Delta E_{SW} = 25$ mV, $f_{SWV} = 60$ Hz: $I_{\text{cap}} = 60$ nA per WE, matching Layer 6's own worked example exactly. Peak faradaic current is 10–100 nA (content spec Sec. 5.3), giving $I_{\text{total, per WE}} \approx$ 70–160 nA.

CE current by multiplexing mode

The CE must carry the *sum* of currents from every WE scanning simultaneously. Two regimes:

Mode	Electrodes active at once	I_{CE}
Sequential (recommended, Sec. IV)	1	70–160 nA
Simultaneous, confirmed design	8 ($2 \times N_{\text{active}} = 4$ pairs, Layer 13 worked example)	560–1280 nA
Simultaneous, original planning	8 (dual-channel cartridge)	560–1280 nA

The two simultaneous-mode rows coincide numerically because both assume 8 electrodes scanning at once; they differ in what those 8 electrodes *are* (4 active/control pairs on a single MB+Fc channel,

vs. a dual-channel MB/Fc + HNE-channel cartridge) rather than in the resulting current, which depends only on electrode count here.

Sizing

Current-density-limited minimum CE area ($j_{CE,\max} = 1$ mA/cm^2 for smooth Au in physiological solution, content spec Sec. 5.3) is trivial in every mode considered (0.016–0.13 mm^2) – several orders of magnitude below any plausible CE footprint. The binding constraint is instead the standard 10×-total-WE-area rule of thumb (content spec Sec. 5.3):

	Total WE area	CE area (10× rule)
Confirmed design (2–8 electrodes)	0.08 cm^2 (8 mm^2)	0.80 cm^2 (80 mm^2)
Original planning (8 electrodes)	0.08 cm^2 (8 mm^2)	0.80 cm^2 (80 mm^2)

Key finding. An 80 mm^2 CE is a substantial fraction of the entire chamber footprint (20 × 5 mm = 100 mm^2, README chamber geometry) – yet the current-density-limited requirement above shows this margin is unnecessary by roughly three orders of magnitude at the currents this device actually draws. The 10× rule of thumb is a conservative default calibrated for CE-limited designs running at or near $j_{CE,\max}$; at $I_{CE} \leq 1.3$ μA into a CE of even 2–3× the total WE area (16–24 mm^2), the resulting current density is still $\sim 10^{-2}$ of $j_{CE,\max}$. A smaller CE (design target: $\geq 3\times$ WE area, not 10×) would leave ample margin while freeing chamber area for the fluidic and WE layout – worth revisiting with the fabrication team rather than defaulting to the generic rule of thumb.

Gas evolution risk

Water oxidation onset: $E^\circ = +1.23$ V vs NHE = +1.03 V vs saturated Ag/AgCl. Comparing to the two scan-window candidates:

Scan window (vs Ag/AgCl)	Upper limit	Margin to O_2 evolution
Confirmed (Layer 7/10)	+0.10 V	933 mV
Original planning (content spec Sec. 5.3)	+0.65 V	383 mV

Both retain comfortable margin, and the confirmed narrower window makes the bubble-formation risk flagged in the content spec essentially moot for the as-designed device – another consequence of the confirmed single-channel design's narrower scan range.

IV. Multiplexing Mode: Shared vs Individual CE

Content spec Sec. 5.4 identifies two options: (a) a single shared CE, current-limited by how many WEs scan simultaneously; (b) an individual CE per WE pair, which eliminates any cross-talk but adds fabrication complexity. No confirmed downstream report specifies simultaneous vs. sequential multiplexing for the fabricated device, so this remains an open design choice rather than a settled fact – unlike the scan window and electrode architecture, which *are* confirmed (Layers 7, 10, 13). Given that a single shared CE under *sequential* multiplexing keeps I_{CE} to 70–160 nA regardless of how many electrode pairs the cartridge ultimately carries (Sec. III), and that individual per-pair CEs

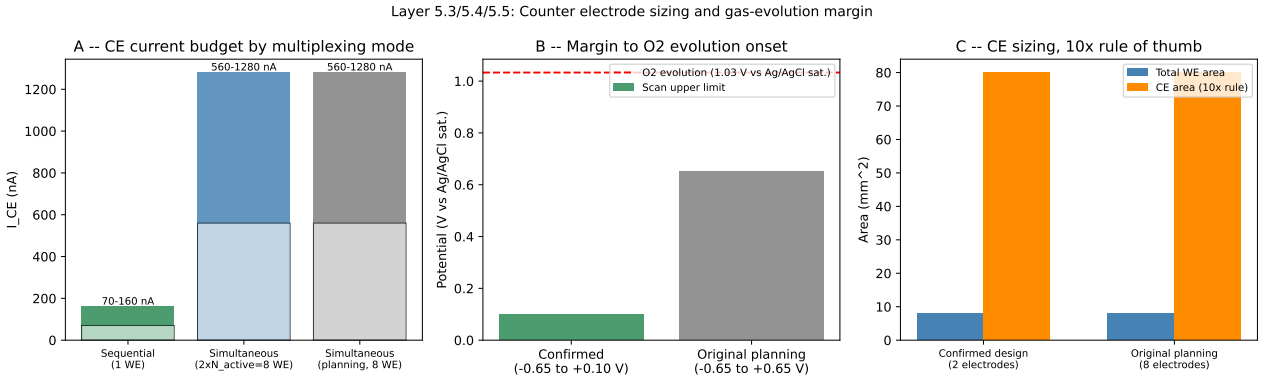


Figure 2: **A**: CE current budget by multiplexing mode. **B**: Margin between each candidate scan window's upper limit and the O_2-evolution onset. **C**: CE area under the 10×-WE-area rule of thumb, for the confirmed and original-planning electrode counts (which coincide here).

add fabrication complexity this design does not need (Sec. 5.4), the recommendation carried forward is: a single shared CE, read sequentially, positioned near the chamber outlet/vent so that any O_2 bubbles (already at very low risk, Sec. III) migrate away from the WE array rather than across it.

V. Solution Resistance and iR Compensation

$$R_s = \rho_{\text{sol}} \cdot \frac{d_{\text{RE-WE}}}{A_{\text{cross}}}, \qquad \rho_{\text{sol}} = 1/\sigma_{\text{sol}} \tag{V.1}$$

For wound fluid ($\sigma_{\text{sol}} = 15$ mS/cm $\Rightarrow \rho_{\text{sol}} = 6.67$ Ω·m), $d_{\text{RE-WE}} = 2$ mm, and a channel cross-section $A_{\text{cross}} = 5\text{ mm} \times 0.375\text{ mm} = 1.875\text{ mm}^2$:

$$R_s = 6.67 \times \frac{2 \times 10^{-3}}{1.875 \times 10^{-6}} \approx 7{,}100\ \Omega \tag{V.2}$$

This looks large for a "wire" resistance, but it is characteristic of a thin-layer cell: the current path's cross-section is small, so even a modest solution conductivity gives a high resistance. What matters is not R_s in isolation but the resulting iR drop at the currents actually drawn:

Current source	Magnitude	iR drop
Capacitive (per WE)	60 nA	0.43 mV
Faradaic (peak, per WE)	100 nA	0.71 mV

Both are $\ll$ the SWV step size ($\Delta E_{\text{step}} = 2$ mV) and $\ll$ the MB peak FWHM (65 mV, Layer 11). **No iR compensation is required** for this device at any expected current: the thin-layer geometry keeps R_s high, but the currents are small enough (nA-scale, from the small electrode areas and low SWV frequency) that the product iR stays two orders of magnitude below the scale that would visibly distort a peak. This confirms the content spec's own conclusion (Sec. 5.5) and provides the closed-form justification to record in the instrument design file.

VI. Cross-Layer Integration Check

Two findings from this report interact with confirmed values elsewhere in the pipeline:

1. **Reference-frame ambiguity (Sec. I).** Layers 7 and 10 quote potentials "vs Ag/AgCl" without stating which convention. The ~ 90 mV difference between conventions is small relative to the confirmed scan window and peak FWHM, so no numeric result in those reports is at risk of being wrong by an amount that matters – but the ambiguity should be resolved explicitly (state "vs on-chip quasi-ref" or "vs sat. Ag/AgCl" in every reported potential) before Layer 11's forward SWV model and Layer 12's peak extraction are combined quantitatively with this report's numbers.

2. **CE oversizing (Sec. III).** The 10×-rule-of-thumb CE area (80 mm^2) would consume most of the 100 mm^2 chamber footprint (README Layer 2 geometry), yet the actual current-density requirement is met by a CE only a few times the WE area. This is a genuine tension between a generic design heuristic and this device's specific (very low) current budget, worth flagging to the hardware design team rather than defaulting to the rule of thumb without checking it against the device's own numbers – exactly the kind of check Layer 5 exists to make explicit.

Both the RE drift budget (Sec. II) and the iR analysis (Sec. V) close out cleanly: neither introduces a practical constraint on the confirmed design, and both numbers (drift < 2 mV; iR drop < 1 mV) are small enough that they can be treated as negligible in the Layer 10/11/12 potential-axis and peak-extraction chain without further correction.

VII. Key Outputs

Output	Value	Feeds
E_{ref} offset from sat. Ag/AgCl	+88 mV (mid; range +87 to +90 mV)	Layer 10, 11 (potential axis)
Max RE drift over assay	< 2 mV (target); < 5 mV acceptable ceiling	Layer 12 (peak-position tolerance)
CE area requirement (current-density-limited)	< 0.13 mm^2 (trivial)	Hardware design spec
CE area, 10×-WE-area rule of thumb	80 mm^2 (flagged as likely oversized)	Hardware design spec
O_2-evolution margin, confirmed scan window	933 mV	Sec. III, CE placement
R_s	$\sim$7.1 kΩ (but $iR < 1$ mV)	Layer 6, 14

VIII. Limitations

1. The chloride activity coefficient $\gamma \approx 0.75$ is a literature Debye-Hückel estimate, not measured for wound fluid. It enters only the E_{ref} offset (Sec. I), where a ± 0.1 change in γ shifts the offset by only a few mV (logarithmic dependence) – not large enough to change any conclusion here, but not confirmed either.

2. The RE drift random-walk model (Sec. II) is calibrated to hit the content spec's own < 2 mV target for a "sound RE design"; it is not derived from a measured Cl^- diffusion coefficient at the RE surface. A real fabricated RE should be characterised by open-circuit-potential logging over a 30-min soak to confirm this budget.

3. $C_{dl} = 2$ $\mu F/cm^2$ (Sec. III) is the Layer 6 mid-range literature value, not a measured value for this SAM composition; the CE current budget scales linearly with it.

4. $j_{CE,\max} = 1$ mA/cm^2 (Sec. III) is a literature estimate for smooth gold in physiological solution, not measured for this electrode's surface finish; roughness (Layer 4, $R_f = 1.1$–1.5) would only increase the effective current-carrying area, making the current-density-limited minimum CE area even more comfortably trivial.

5. The reference-frame ambiguity identified in Sec. I (Cross-Layer Integration Check, item 1) is not resolved by this report – it requires an explicit convention statement from whoever generated Layer 10/11's potential values.

6. Simultaneous-vs-sequential multiplexing (Sec. IV) is a recommendation, not a confirmed design decision; no downstream report specifies the actual SWV acquisition sequence for the fabricated cartridge.

7. The proposed literature source for RE stability in biological fluids (content spec Sec. "Key literature to find": "White & Plaxco review") was not identified as a paper specifically addressing on-chip Ag/AgCl RE stability; White RJ and Plaxco KW's confirmed 2010 *Anal. Chem.* paper (already cited in Layer 13/15) addresses ratiometric probe-flexibility design, not RE stability. Polk et al. (2006) is the operative confirmed source for on-chip RE fabrication and stability; the RE-stability literature gap should be revisited with a targeted search rather than assumed closed.

References

Papers used directly in this report; consistent with citations already confirmed in Layers 7, 10, and 13.

1. Bard AJ, Faulkner LR (2001). *Electrochemical Methods: Fundamentals and Applications*, 2nd ed. Wiley, New York. – standard Ag/AgCl and saturated-KCl reference potentials (Sec. I, Eq. I.1), and CE design principles (Chapter 15, content spec "Key literature to find").

2. Polk BJ, Stelzenmuller A, Mijares GI, MacCrehan WA, Gaitan M (2006). Ag/AgCl microelectrodes with improved stability for microfluidics. *Sens. Actuators B Chem.* 114(1):239–247. – electroplated-silver/chloridised thin-film Ag/AgCl microreference electrode fabrication and stability characterisation for microfluidic devices; operative source for the on-chip RE construction assumed throughout Sec. I–II.

3. Pheeney CG, Barton JK (2012). DNA electrochemistry with tethered methylene blue. *Langmuir* 28:7063–7070. DOI: 10.1021/la300566x. – $E^{\circ\prime}_{\mathrm{MB}} = -290$ mV vs Ag/AgCl at pH 7 (confirmed, Layer 10), corrected to the on-chip quasi-reference scale in Sec. I.